PULLBACK TO A BIGGER BIG SITE PRESERVES REPRESENTABILITY AND THE ALGEBRAIC AND DELIGNE-MUMFORD STACK PROPERTIES

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Editorial note. The source handles (1), (2), (4) by 'translate into the same kind of presentation'; the mechanism is exactly the (3) case and each case is carried as an explicit obligation.

Allowed standard prerequisites (editorial).

- Presentation existence. lemma-stack-presentation (carded L26691). Writes X = [U/R] for a smooth groupoid in algebraic spaces.
- Change of big site for quotient stacks. Groupoids in Spaces, lemma quotient-stack-change-big-site (cross-chapter, named). f^{-1}[U/R] = [f^{-1}U/f^{-1}R].
- Preservation of the groupoid type. the evident functoriality of f^{-1} on morphisms of algebraic spaces (structural). f^{-1}U, f^{-1}R, f^{-1}s, f^{-1}t, f^{-1}c form a smooth (resp. etale) groupoid again.

Source reference. Original lemma 04T5: sources/stacks-algebraic/pinned-original-source.tex, source lines 2494–2494. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04WX: focused-seeds/geometry/S9-alg-change-big-site/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 022I: focused-seeds/geometry/S9-alg-change-big-site/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Suppose given big sites Sch_{fppf} and Sch'_{fppf}. Assume that Sch_{fppf} is contained in Sch'_{fppf}, see Topologies, Section 022I. Let S be an object of Sch_{fppf}. Let $f : (Sch'/S)_{fppf} \to (Sch/S)_{fppf}$ the morphism of sites corresponding to the inclusion functor $u : (Sch/S)_{fppf} \to (Sch'/S)_{fppf}$. Let $\mathcal{X}$ be a stack in groupoids over $(Sch/S)_{fppf}$.*

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *if $\mathcal{X}$ is representable by some $X \in \mathrm{Ob}((Sch/S)_{fppf})$, then $f^{-1}\mathcal{X}$ is representable too, in fact it is representable by the same scheme X, now viewed as an object of $(Sch'/S)_{fppf}$,*
(2) *if $\mathcal{X}$ is representable by $F \in Sh((Sch/S)_{fppf})$ which is an algebraic space, then $f^{-1}\mathcal{X}$ is representable by the algebraic space $f^{-1}F$,*
(3) *if $\mathcal{X}$ is an algebraic stack, then $f^{-1}\mathcal{X}$ is an algebraic stack, and*
(4) *if $\mathcal{X}$ is a Deligne-Mumford stack, then $f^{-1}\mathcal{X}$ is a Deligne-Mumford stack too.*

Proof. Let us prove (3). By Lemma 04T5 we may write $\mathcal{X} = [U/R]$ for some smooth groupoid in algebraic spaces (U, R, s, t, c). By Groupoids in Spaces, Lemma 04WX we see that $f^{-1}[U/R] = [f^{-1}U/f^{-1}R]$. Of course $(f^{-1}U, f^{-1}R, f^{-1}s, f^{-1}t, f^{-1}c)$ is a smooth groupoid in algebraic spaces too. Hence (3) is proved.

Now the other cases (1), (2), (4) each mean that $\mathcal{X}$ has a presentation $[U/R]$ of a particular kind, and hence translate into the same kind of presentation for $f^{-1}\mathcal{X} = [f^{-1}U/f^{-1}R]$. Whence the lemma is proved. □